

Advanced Statistical Methods II

Lecture V: EM for Normal Mixtures

Monitirtha Dey



BSDS Third Year Students
Indian Statistical Institute

First Semester, AY 2026–2027

Outline of Today's Lecture

Recap and objective

Why normal mixtures?

Observed and complete data likelihoods

E-step: responsibilities

M-step: update formulas

Practical issues

Key takeaway messages

A problem

Summary of Lecture IV



In Lecture IV, we introduced EM through the incomplete-data likelihood viewpoint.

- ▶ Observed data: Y .
- ▶ Missing or latent data: Z .
- ▶ Complete-data log-likelihood:

$$\ell_c(\theta; Y, Z) = \log p(Y, Z \mid \theta).$$

- ▶ Observed-data log-likelihood:

$$\ell(\theta; Y) = \log p(Y \mid \theta) = \log \sum_z p(Y, z \mid \theta),$$

with an integral instead of a sum when Z is continuous.

EM replaces the missing complete-data information by its conditional expectation under the current parameter value.

See [Dempster et al. \(1977\)](#); [McLachlan and Krishnan \(2008\)](#), Chs. 1–3.

The EM template from Lecture IV



At iteration t , define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y].$$

E-step

Compute the conditional distribution of the missing information:

$$p(Z \mid Y, \theta^{(t)}),$$

and hence compute $Q(\theta \mid \theta^{(t)})$.

M-step

Update

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)}).$$

How do we fit a model when the population is made of several hidden subpopulations, but we do not observe the subpopulation labels?

For mixtures, EM is a standard way to turn hidden subgroup membership into an estimable likelihood problem.

See [McLachlan and Peel \(2000\)](#); [McLachlan and Krishnan \(2008\)](#).

For each observation $i = 1, \dots, n$:

- $$Z_i \in \{1, \dots, K\}, \quad \mathbb{P}(Z_i = k) = \pi_k.$$

- $$Y_i \mid Z_i = k \sim f_k(y; \theta_k).$$

$$f(y; \theta) = \sum_{k=1}^K \pi_k f_k(y; \theta_k), \quad \pi_k > 0, \quad \sum_{k=1}^K \pi_k = 1.$$

Mixture density: intuition



For $K = 2$,

$$f(y; \theta) = \pi \phi(y; \mu_1, \sigma_1^2) + (1 - \pi) \phi(y; \mu_2, \sigma_2^2).$$

- ▶ π controls the relative size of component 1.
- ▶ μ_1, μ_2 control the centers of the two components.
- ▶ σ_1^2, σ_2^2 control the within-component spreads.

A mixture can produce skewness, multimodality and heavy-tailed-looking shapes even when every component is normal.

The word *mixture* means that each observation has one hidden component label, but we do not observe it.

We follow the same notation as Lecture IV.

Symbol	Meaning
Y_i	observed response for unit i
Z_i	unobserved component label
Z_{ik}	indicator $\mathbb{I}(Z_i = k)$
θ	full parameter vector
$\theta^{(t)}$	current parameter value in EM iteration t
$\tau_{ik}^{(t)}$	posterior responsibility of component k for observation i

The main new object today is

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i),$$

called the **responsibility**.

The observed-data likelihood



We observe only $y_1, \dots, y_n$, not $z_1, \dots, z_n$.

The observed-data likelihood is

$$L(\theta) = \prod_{i=1}^n \left\{ \sum_{k=1}^K \pi_k \phi(y_i; \mu_k, \sigma_k^2) \right\}.$$

Therefore the observed-data log-likelihood is

$$\ell(\theta) = \sum_{i=1}^n \log \left\{ \sum_{k=1}^K \pi_k \phi(y_i; \mu_k, \sigma_k^2) \right\}.$$

The log of a sum is the source of computational difficulty. It prevents the likelihood from separating cleanly by component.

Why direct maximization is awkward



If the component labels were known, each observation would belong to one normal sample.

But without labels,

$$\ell(\theta) = \sum_{i=1}^n \log \{ \pi_1 \phi_{i1} + \cdots + \pi_K \phi_{iK} \}.$$

This creates several difficulties:

- ▶ the log-likelihood is usually non-concave;
- ▶ there may be multiple local maxima;
- ▶ closed-form maximizers are generally not available from the observed likelihood directly.

Introduce latent allocation indicators



Define

$$Z_{ik} = \mathbb{I}(Z_i = k), \quad k = 1, \dots, K.$$

For each i , exactly one component label is active:

$$Z_{ik} \in \{0, 1\}, \quad \sum_{k=1}^K Z_{ik} = 1.$$

The complete data are

$$\{(y_i, z_{i1}, \dots, z_{iK}) : i = 1, \dots, n\}.$$

The missing data are missing **classification labels**.

$$\prod_{k=1}^K [\pi_k \phi(y_i; \mu_k, \sigma_k^2)]^{Z_{ik}}.$$
$$L_c(\theta) = \prod_{i=1}^n \prod_{k=1}^K [\pi_k \phi(y_i; \mu_k, \sigma_k^2)]^{Z_{ik}}.$$
$$\ell_c(\theta) = \sum_{i=1}^n \sum_{k=1}^K Z_{ik} \{ \log \pi_k + \log \phi(y_i; \mu_k, \sigma_k^2) \}.$$

Complete-data log-likelihood for normal components



For the univariate normal density,

$$\log \phi(y_i; \mu_k, \sigma_k^2) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2}.$$

Ignoring the constant $-\frac{n}{2} \log(2\pi)$,

$$\ell_c(\theta) = \sum_{i=1}^n \sum_{k=1}^K Z_{ik} \left[\log \pi_k - \frac{1}{2} \log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2} \right] + \text{constant}.$$

This expression is easy to maximize if we know the labels Z_{ik} . EM will replace Z_{ik} by its conditional expectation.

If labels were known: complete-data MLEs



If Z_{ik} 's were observed, define

$$n_k = \sum_{i=1}^n Z_{ik}.$$

The complete-data MLEs would be

$$\hat{\pi}_k = \frac{n_k}{n},$$

$$\hat{\mu}_k = \frac{\sum_{i=1}^n Z_{ik} y_i}{\sum_{i=1}^n Z_{ik}},$$

$$\hat{\sigma}_k^2 = \frac{\sum_{i=1}^n Z_{ik} (y_i - \hat{\mu}_k)^2}{\sum_{i=1}^n Z_{ik}}.$$

The M-step will have exactly the same form, except that hard labels Z_{ik} will be replaced by soft labels $\tau_{ik}^{(t)}$.

Derivation I: mixing proportions



The part of ℓ_c involving $\pi_1, \dots, \pi_K$ is

$$\ell_{c,\pi} = \sum_{k=1}^K n_k \log \pi_k, \quad n_k = \sum_{i=1}^n Z_{ik},$$

subject to the constraint

$$\sum_{k=1}^K \pi_k = 1.$$

Use the Lagrangian

$$\begin{aligned} \mathcal{L}(\pi_1, \dots, \pi_K, \lambda) &= \sum_{k=1}^K n_k \log \pi_k + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right). \\ \implies \frac{\partial \mathcal{L}}{\partial \pi_k} &= \frac{n_k}{\pi_k} + \lambda = 0 \implies \pi_k = -\frac{n_k}{\lambda}. \end{aligned}$$

Summing over k ,

$$1 = \sum_{k=1}^K \pi_k = -\frac{1}{\lambda} \sum_{k=1}^K n_k = -\frac{n}{\lambda},$$

Derivation II: component means



For a fixed component k , the terms involving μ_k are

$$-\frac{1}{2\sigma_k^2} \sum_{i=1}^n Z_{ik}(y_i - \mu_k)^2.$$

Thus maximizing with respect to μ_k is equivalent to minimizing

$$S_k(\mu_k) = \sum_{i=1}^n Z_{ik}(y_i - \mu_k)^2.$$

Differentiating and setting the derivative equal to zero gives:

$$\sum_{i=1}^n Z_{ik}(y_i - \hat{\mu}_k) = 0.$$

Therefore

$$\hat{\mu}_k = \frac{\sum_{i=1}^n Z_{ik}y_i}{\sum_{i=1}^n Z_{ik}} = \frac{1}{n_k} \sum_{i:Z_i=k} y_i.$$

Derivation III: component variances



Let $v_k = \sigma_k^2$. For a fixed k , after substituting $\hat{\mu}_k$, focus on

$$\begin{aligned} g(v_k) &= -\frac{n_k}{2} \log v_k - \frac{1}{2v_k} \sum_{i=1}^n Z_{ik}(y_i - \hat{\mu}_k)^2 \\ &= -\frac{n_k}{2} \log v_k - \frac{A_k}{2v_k} \\ \implies g'(v_k) &= -\frac{n_k}{2v_k} + \frac{A_k}{2v_k^2}. \end{aligned}$$

Set $g'(v_k) = 0$:

$$-n_k v_k + A_k = 0 \implies \boxed{\hat{\sigma}_k^2 = \hat{v}_k = \frac{A_k}{n_k}} = \frac{\sum_{i=1}^n Z_{ik}(y_i - \hat{\mu}_k)^2}{\sum_{i=1}^n Z_{ik}}.$$

What must be computed in the E-step?



The E-step computes

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y].$$

In the complete-data log-likelihood, the only random missing quantities are the indicators Z_{ik} .

So we need

$$\mathbb{E}_{\theta^{(t)}}[Z_{ik} \mid Y_i = y_i].$$

Since Z_{ik} is an indicator,

$$\mathbb{E}_{\theta^{(t)}}[Z_{ik} \mid Y_i = y_i] = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i).$$

This conditional probability is called the **responsibility**:

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i).$$

Derivation of the responsibility formula



By Bayes' rule,

$$\mathbb{P}(Z_i = k \mid Y_i = y_i, \theta^{(t)}) = \frac{\mathbb{P}(Z_i = k \mid \theta^{(t)})f(y_i \mid Z_i = k, \theta^{(t)})}{f(y_i \mid \theta^{(t)})}.$$

Under the mixture model,

$$\mathbb{P}(Z_i = k \mid \theta^{(t)}) = \pi_k^{(t)},$$

$$f(y_i \mid Z_i = k, \theta^{(t)}) = \phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)}),$$

and

$$f(y_i \mid \theta^{(t)}) = \sum_{\ell=1}^K \pi_{\ell}^{(t)} \phi(y_i; \mu_{\ell}^{(t)}, (\sigma_{\ell}^2)^{(t)}).$$

E-step formula



Therefore,

$$\tau_{ik}^{(t)} = \frac{\pi_k^{(t)} \phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)})}{\sum_{\ell=1}^K \pi_{\ell}^{(t)} \phi(y_i; \mu_{\ell}^{(t)}, (\sigma_{\ell}^2)^{(t)})}$$

for $i = 1, \dots, n$ and $k = 1, \dots, K$.

$\tau_{ik}^{(t)}$ measures how much component k is responsible for observation i , given the current parameter estimates.

For each observation,

$$0 \leq \tau_{ik}^{(t)} \leq 1, \quad \sum_{k=1}^K \tau_{ik}^{(t)} = 1.$$

$$Y_i \longrightarrow \text{one component only.}$$
$$Y_i \longrightarrow (\tau_{i1}^{(t)}, \dots, \tau_{iK}^{(t)}).$$

This is why EM for mixtures is often described as **soft clustering**.

The Q-function for normal mixtures



Replace Z_{ik} in the complete-data log-likelihood by $\tau_{ik}^{(t)}$.

Thus,

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \tau_{ik}^{(t)} \left[\log \pi_k - \frac{1}{2} \log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2} \right] + \text{constant}.$$

The E-step has converted the missing-label problem into a weighted normal likelihood problem.

The weights are not arbitrary; they are posterior probabilities under the current parameters.

M-step overview



The M-step maximizes

$$Q(\theta \mid \theta^{(t)})$$

with respect to

$$\pi_1, \dots, \pi_K, \quad \mu_1, \dots, \mu_K, \quad \sigma_1^2, \dots, \sigma_K^2.$$

The resulting updates are:

$$\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \tau_{ik}^{(t)},$$

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} y_i}{\sum_{i=1}^n \tau_{ik}^{(t)}},$$

$$(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^n \tau_{ik}^{(t)}}.$$

Deriving the update for mixing proportions

The part of Q involving π_k 's is

$$Q_\pi = \sum_{i=1}^n \sum_{k=1}^K \tau_{ik}^{(t)} \log \pi_k, \quad \sum_{k=1}^K \pi_k = 1.$$

Use a Lagrange multiplier:

$$\mathcal{L} = Q_\pi + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right).$$

Differentiating,

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)}}{\pi_k} + \lambda = 0.$$

Thus,

$$\pi_k = -\frac{\sum_i \tau_{ik}^{(t)}}{\lambda}.$$

Completing the derivation for π_k



Use the constraint $\sum_k \pi_k = 1$:

$$\sum_{k=1}^K \left(-\frac{\sum_i \tau_{ik}^{(t)}}{\lambda} \right) = 1.$$

Since

$$\sum_{k=1}^K \sum_{i=1}^n \tau_{ik}^{(t)} = \sum_{i=1}^n \sum_{k=1}^K \tau_{ik}^{(t)} = n,$$

we get $-n/\lambda = 1$, hence $\lambda = -n$.

Therefore,

$$\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \tau_{ik}^{(t)}$$

The updated mixing proportion is the average responsibility of component k .

Deriving the update for μ_k



For a fixed component k , the part of Q involving μ_k is

$$-\frac{1}{2\sigma_k^2} \sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k)^2.$$

Maximizing this is equivalent to minimizing

$$\sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k)^2.$$

Differentiate with respect to μ_k :

$$\frac{\partial}{\partial \mu_k} \sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k)^2 = -2 \sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k).$$

Setting this equal to zero gives

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} y_i}{\sum_{i=1}^n \tau_{ik}^{(t)}}$$

Deriving the update for σ_k^2



For a fixed component k , the part involving σ_k^2 is

$$-\frac{1}{2} \sum_{i=1}^n \tau_{ik}^{(t)} \left[\log \sigma_k^2 + \frac{(y_i - \mu_k)^2}{\sigma_k^2} \right].$$

Let $v_k = \sigma_k^2$. Differentiate with respect to v_k :

$$\frac{\partial Q}{\partial v_k} = -\frac{1}{2} \sum_{i=1}^n \tau_{ik}^{(t)} \left[\frac{1}{v_k} - \frac{(y_i - \mu_k)^2}{v_k^2} \right].$$

Set the derivative equal to zero:

$$\sum_{i=1}^n \tau_{ik}^{(t)} v_k = \sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k)^2.$$

Variance update



Using the updated mean $\mu_k^{(t+1)}$, we get

$$(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^n \tau_{ik}^{(t)}}$$

The M-step is simply a weighted version of the usual MLEs for normal data.

Quantity	Complete-data MLE	EM M-step
sample size in component k	$\sum_i Z_{ik}$	$\sum_i \tau_{ik}^{(t)}$
component mean	hard average	weighted average
component variance	hard variance	weighted variance

The EM algorithm for univariate normal mixtures



Algorithm

Choose starting values

$$\theta^{(0)} = (\pi_k^{(0)}, \mu_k^{(0)}, (\sigma_k^2)^{(0)})_{k=1}^K.$$

For $t = 0, 1, 2, \dots$:

1. **E-step:** compute responsibilities $\tau_{ik}^{(t)}$.
2. **M-step:** update π_k, μ_k, σ_k^2 using weighted MLE formulas.
3. Compute the observed log-likelihood

$$\ell(\theta^{(t)}) = \sum_i \log \left\{ \sum_k \pi_k^{(t)} \phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)}) \right\}.$$

4. Stop when the parameter changes or likelihood improvement is small.

Starting values can be chosen using:

- See [McLachlan and Peel \(2000\)](#), Chs. 2–3.

Choosing the number of components



So far, we have treated K as known.

In practice, K may be chosen using:

- ▶ scientific knowledge about the number of subpopulations;
- ▶ likelihood-based criteria such as AIC or BIC;
- ▶ validation and interpretability.

A larger K can fit the data better, but may overfit or produce components with no meaningful interpretation.

- $$\tau_{ik}^{(t)} = \mathbb{P}_{\theta(t)}(Z_i = k \mid Y_i = y_i).$$

- ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ▶ ↺ 🔍 ↻

A Question



Suppose it is known

$$\mu_1 = \mu_2 = \mu \quad \& \quad \sigma_1^2 = \sigma_2^2 = \sigma^2.$$

Then what would be the E-step and the M-step?

Bivariate Normal with one missing value



Suppose

$$(x_1, y_1), (x_2, y_2), \dots, (x_{n-1}, y_{n-1}), (x_n, *)$$

are treated as observations from a Bivariate Normal distribution, where $*$ denotes a missing value and

$$x_n = \bar{x}_{n-1} = \frac{1}{n-1} \sum_{i=1}^{n-1} x_i.$$

Use the MLEs based on the first $n - 1$ complete observations as initial values.

Show that the corresponding EM algorithm converges in two cycles of iteration.

Notation and initial values

Let $m = n - 1$. From the first m complete observations define the MLEs

$$\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i, \quad \bar{y} = \frac{1}{m} \sum_{i=1}^m y_i,$$

$$s_{xx} = \frac{1}{m} \sum_{i=1}^m (x_i - \bar{x})^2, \quad s_{yy} = \frac{1}{m} \sum_{i=1}^m (y_i - \bar{y})^2, \quad s_{xy} = \frac{1}{m} \sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y}).$$

The initial values are therefore

$$\mu_x^{(0)} = \bar{x}, \quad \mu_y^{(0)} = \bar{y}, \quad \sigma_{xx}^{(0)} = s_{xx}, \quad \sigma_{yy}^{(0)} = s_{yy}, \quad \sigma_{xy}^{(0)} = s_{xy}.$$

But the problem states that

$$X_n = \bar{X}.$$

E-step in the first cycle

For a bivariate normal distribution,

$$Y \mid X = x \sim N \left(\mu_y + \frac{\sigma_{xy}}{\sigma_{xx}}(x - \mu_x), \sigma_{yy} - \frac{\sigma_{xy}^2}{\sigma_{xx}} \right).$$

At the initial value, $x_n = \mu_x^{(0)} = \bar{x}$. Hence

$$\hat{y}_n^{(1)} = \mathbb{E}_{\theta^{(0)}}(Y_n \mid X_n = x_n) = \mu_y^{(0)} = \bar{y}.$$

Also,

$$\mathbb{E}_{\theta^{(0)}}(Y_n^2 \mid X_n = x_n) = \bar{y}^2 + \tau^2,$$

where

$$\tau^2 = s_{yy} - \frac{s_{xy}^2}{s_{xx}}.$$

M-step in the first cycle

Using the expected completed data,

$$\mu_x^{(1)} = \frac{m\bar{x} + x_n}{n} = \bar{x}, \quad \mu_y^{(1)} = \frac{m\bar{y} + \bar{y}}{n} = \bar{y}.$$

For the covariance terms,

$$\sigma_{xx}^{(1)} = \frac{1}{n} \left\{ \sum_{i=1}^m (x_i - \bar{x})^2 + (x_n - \bar{x})^2 \right\} = \frac{m}{n} s_{xx},$$

$$\sigma_{xy}^{(1)} = \frac{1}{n} \left\{ \sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y}) + (x_n - \bar{x}) \mathbb{E}(Y_n - \bar{y} \mid x_n) \right\} = \frac{m}{n} s_{xy}.$$

For the Y -variance,

$$\sigma_{yy}^{(1)} = \frac{1}{n} \{ ms_{yy} + \mathbb{E}[(Y_n - \bar{y})^2 | x_n] \} = \frac{1}{n} \left\{ ms_{yy} + s_{yy} - \frac{s_{xy}^2}{s_{xx}} \right\}.$$

Thus

$$\sigma_{yy}^{(1)} = s_{yy} - \frac{1}{n} \frac{s_{xy}^2}{s_{xx}}.$$

Why the next cycle changes nothing



In the second E-step, still

$$x_n = \mu_x^{(1)} = \bar{x},$$

so the conditional mean remains

$$\mathbb{E}_{\theta^{(1)}}(Y_n \mid X_n = x_n) = \mu_y^{(1)} = \bar{y}.$$

The conditional variance under $\theta^{(1)}$ is

$$\sigma_{yy}^{(1)} - \frac{(\sigma_{xy}^{(1)})^2}{\sigma_{xx}^{(1)}}.$$

To prove convergence, it is enough to show that this conditional variance is also the same as in the first E-step.

Why the next cycle changes nothing, continued



Substitute the first-cycle values:

$$\left(s_{yy} - \frac{1}{n} \frac{s_{xy}^2}{s_{xx}} \right) - \frac{\left(\frac{m}{n} s_{xy} \right)^2}{\frac{m}{n} s_{xx}} = \left(s_{yy} - \frac{1}{n} \frac{s_{xy}^2}{s_{xx}} \right) - \frac{m}{n} \frac{s_{xy}^2}{s_{xx}}.$$

Since $m = n - 1$,

$$\frac{1}{n} + \frac{m}{n} = 1.$$

Therefore

$$\sigma_{yy}^{(1)} - \frac{(\sigma_{xy}^{(1)})^2}{\sigma_{xx}^{(1)}} = s_{yy} - \frac{s_{xy}^2}{s_{xx}} = \tau^2.$$

The second E-step gives exactly the same first and second conditional moments as the first E-step. Hence the next M-step reproduces $\theta^{(1)}$.

Problem 1: conclusion



The EM sequence is

$$\theta^{(0)} \longrightarrow \theta^{(1)} \longrightarrow \theta^{(2)} = \theta^{(1)}.$$

Thus the algorithm has reached a fixed point after the first update; equivalently, after checking the second cycle, it has converged in two cycles of iteration.

The result depends crucially on $x_n = \bar{x}$. Because the missing observation is located at the current mean of X , the conditional mean imputation for Y_n remains $\bar{y}$, and the conditional variance stabilizes after the first EM update.

Do not replace Y_n only by its conditional mean when estimating variances. EM uses the conditional second moment, which includes the conditional variance term.

- Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1), 1–38.
- McLachlan, G. J. and Krishnan, T. (2008). *The EM Algorithm and Extensions*. 2nd ed., Wiley.
- McLachlan, G. J. and Peel, D. (2000). *Finite Mixture Models*. Wiley.